

torch.lu_unpack#

https://pytorch.org/docs/stable/generated/torch.lu_unpack.html#torch.lu_unpack

Description:

Unpacks the LU decomposition returned by `lu_factor()` into the P, L, U matrices. `lu()` returns the matrices from the LU decomposition. Its gradient formula is more efficient than that of doing `lu_factor()` followed by `lu_unpack()`. LU_data (Tensor) – the packed LU factorization data LU_pivots (Tensor) – the packed LU factorization pivots

Function Signature

```
torch.lu_unpack(LU_data, LU_pivots, unpack_data=True,
unpack_pivots=True, *, out=None)#
```

Parameters

Keyword Arguments

Returns

Parameters

Keyword

out (tuple, optional) – output tuple of three tensors. Ignored if None.

Returns:

A namedtuple (P, L, U)

Code Examples

```
>>> A = torch.randn(2, 3, 3)
>>> LU, pivots = torch.linalg.lu_factor(A)
>>> P, L, U = torch.lu_unpack(LU, pivots)
>>> # We can recover A from the factorization
>>> A_ = P @ L @ U
>>> torch.allclose(A, A_)
True

>>> # LU factorization of a rectangular matrix:
>>> A = torch.randn(2, 3, 2)
>>> LU, pivots = torch.linalg.lu_factor(A)
>>> P, L, U = torch.lu_unpack(LU, pivots)
>>> # P, L, U are the same as returned by linalg.lu
>>> P_, L_, U_ = torch.linalg.lu(A)
>>> torch.allclose(P, P_) and torch.allclose(L, L_) and
torch.allclose(U, U_)
True
```

Notes & Warnings

NOTE

See also `lu()` returns the matrices from the LU decomposition. Its gradient formula is more efficient than that of doing `lu_factor()` followed by `lu_unpack()`.

Detailed Documentation

Documentation

Unpacks the LU decomposition returned by `lu_factor()` into the P, L, U matrices.

`lu()` returns the matrices from the LU decomposition. Its gradient formula is more efficient than that of doing `lu_factor()` followed by `lu_unpack()`.

LU_data (Tensor) – the packed LU factorization data

LU_pivots (Tensor) – the packed LU factorization pivots

unpack_data (bool) – flag indicating if the data should be unpacked. If False, then the returned L and U are empty tensors. Default: True

unpack_pivots (bool) – flag indicating if the pivots should be unpacked into a permutation matrix P. If False, then the returned P is an empty tensor. Default: True

out (tuple, optional) – output tuple of three tensors. Ignored if None.

A namedtuple (P, L, U)